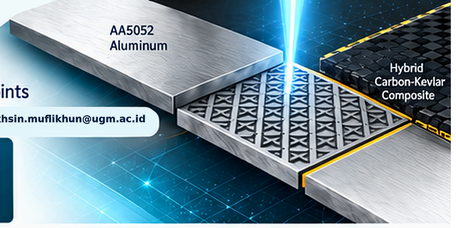
BIO-INSPIRED GRADED-X LASER TEXTURING FOR STRONGER ALUMINUM-COMPOSITE JOINTS

Enthesis-informed modular graded-X laser texturing for dissimilar AA5052 / hybrid carbon–Kevlar adhesive joints

Researcher: Rahmad Kuncoro Adi | Supervisors: Prof. Gesang Nugroho; M. Akhsin Muflikhun, Ph.D. | DTMI, Universitas Gadjah Mada | akhsin.muflikhun@ugm.ac.id



15.73 MPa — L4-60 graded-X

96.7% — strength retained vs full-overlap L1

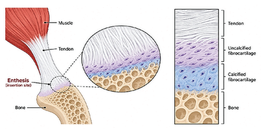
271.9% — strength gain vs untextured control C0

1 WHY THIS STUDY?

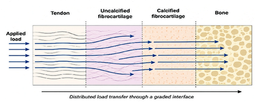
- Lightweight structures often join aluminum to fiber-reinforced composites.
- Single-lap joints suffer from strong peel-shear concentration near overlap ends.
- Surface texturing may improve adhesion, but placement and architecture matter.
- This study investigates whether targeted laser texturing near overlap ends can achieve high joint strength with less treated length.

2 ENTHESIS-INFORMED CONCEPT

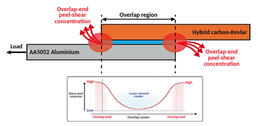
(a) muscle-tendon-enthesis-bone anatomy



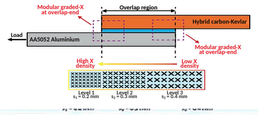
(b) the functional principle at the entheses



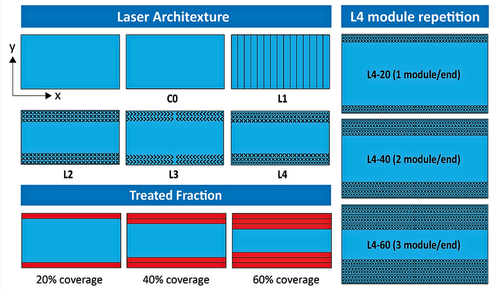
(c) engineering problem: overlap-end localization



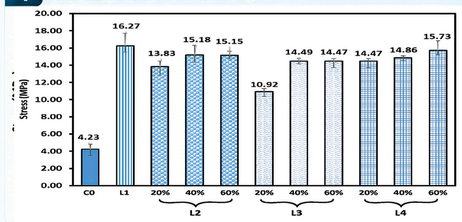
(d) engineering translation: modular graded-X architexture at overlap-ends



3 LASER ARCHITECTURE / EXPERIMENTAL DESIGN



4 MECHANICAL PERFORMANCE

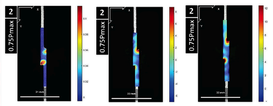


L4-60 reached 15.73 MPa, retaining 96.7% of the L1 mean within a nominal 60% treatment envelope.

No statistical superiority of L4-60 over all localized alternatives was established.

5 DEFORMATION EVIDENCE (DIC)

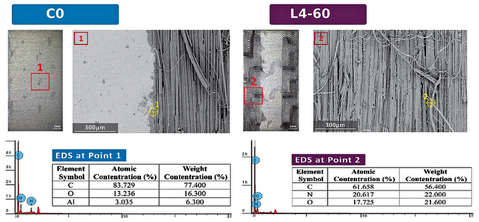
Vertical strain field at 0.75 P_{max}



	C0	L1	L4-60
ε_{SS}	0.09560	0.00663	0.00386
Kε	3.468	1.907	2.429
f_{50}	19.24%	46.88%	31.19%

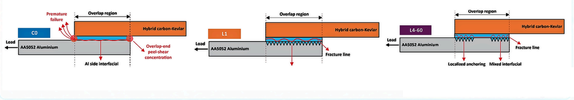
Textured joints show reduced strain localization relative to the untextured control.

6 FRACTURE / MICROSTRUCTURAL EVIDENCE



C0: interfacial debonding. L4-60: adhesive, matrix and fiber damage.

7 INTERPRETATION / MECHANISM



8 KEY TAAKAWAYS

1. Laser texturing substantially increases joint strength compared with the untextured control.
2. L4-60 achieved 15.73 MPa and retained 96.7% of L1 within a 60% treatment envelope.
3. DIC results show reduced strain localization for textured joints.
4. Fracture and SEM/EDS analyses indicate a transition from simple interfacial debonding (C0) to mixed interfacial failure with localized anchoring (L4-60).

Limitation: The CZM model was calibrated for the baseline C0 joint; three-dimensional topography, fatigue, and durability remain for future work.